(iii) SSS (Side-Side-Side) Similarity

Two triangles are similar if their corresponding sides are in the same ratio. That is, if $\frac{AB}{PQ}=\frac{AC}{PR}=\frac{BC}{QR}$, then $\Delta ABC \sim \Delta PQR$.

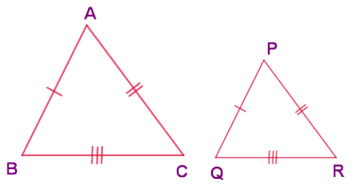


Fig. 5.14

(iv) RHS (Right Angle – Hypotenuse – Side) Similarity

Two right triangles are similar if the hypotenuse and a leg of one triangle are respectively proportional to the hypotenuse and a leg of the other triangle. That is, if $\angle B=\angle Q=90^\circ$ and $\frac{AC}{PR}=\frac{BC}{QR}$ then, $\Delta ABC \sim \Delta PQR$.

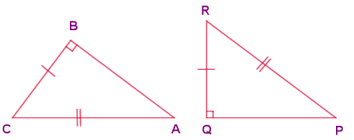
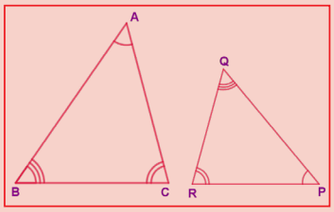


Fig. 5.15

DO YOU KNOW?

If $\Delta ABC \sim \Delta PQR$, then the corresponding sides to AB, BC and AC of ΔABC are PQ, QR and PR respectively and the corresponding angles to A, B and C are P, Q and R respectively. Naming a triangle has a significance. For example, if $\Delta ABC \sim \Delta PQR$ then, ΔBAC is not similar to ΔPQR.



Example 5.5

In the Fig. 5.16, if $\Delta PQR \sim \Delta XYZ$, find a and b.

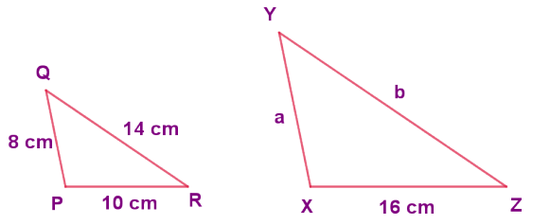


Fig. 5.16

Solution:

Given that $\Delta PQR \sim \Delta XYZ$

$\therefore$ Their corresponding sides are proportional.

$$\Rightarrow \frac{PQ}{XY}=\frac{QR}{YZ}=\frac{PR}{XZ}$$

$$\Rightarrow \frac{8}{a}=\frac{14}{b}=\frac{10}{16}$$

$$\Rightarrow \frac{8}{a}=\frac{10}{16}$$

$$\Rightarrow a=\frac{8\times 16}{10}=\frac{128}{10}$$

$$a = 12.8\ cm$$

Also, $$\frac{14}{b}=\frac{10}{16}$$

$$\Rightarrow b=\frac{14\times 16}{10}=\frac{224}{10}$$

$$\therefore\ b = 22.4\ cm$$

Note

- All circles and squares are similar to each other.
- Not all rectangles need to be similar always.
- If two angles are both congruent and supplementary then, they are right angles.
- All congruent triangles are similar.
- The symbol $\sim$ is used to denote similarity.

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